On a March visit to Chicago, I got a jolt. I gave a seminar describing everything I understood about wormholes and time machines; and after the seminar, Geroch and Wald asked me (in effect), "*Won't a wormhole be automatically destroyed whenever an advanced civilization tries to convert it into a time machine?*"

Why? How? I wanted to know. They explained. Translated into the language of the Carolee-and-me story, their explanation was the following: Imagine that Carolee is zooming back to Earth with one wormhole mouth in her spacecraft and I am sitting at home on Earth with the other. Shortly after she turns around and begins zooming home, it suddenly becomes possible for radiation (electromagnetic waves) to use the wormhole for time travel: Any random bit of radiation that leaves our home in Pasadena traveling at the speed of light toward her spacecraft can arrive at the spacecraft after 5 years' time (as seen on Earth), enter the wormhole mouth there, travel back in time by 5 years (as seen on Earth), and emerge from the mouth on Earth at precisely the same moment as it started its trip. The radiation piles right on top of its previous self, not just in space but in spacetime, doubling its strength. What's more, during the trip each quantum of radiation (each photon) got boosted in energy due to the relative motion of the wormhole mouths (a "Doppler-shift" boost).

After the radiation's next trip out to the spacecraft then back through the wormhole, it again returns at the same time as it left and again piles up on itself, again with a Doppler-boosted energy. Again and again this happens, making the beam of radiation infinitely strong (Figure 14.8a).